

CSA0404 – OPERATING SYSTEMS – LAB PROGRAMS

EXPERIMENT 20

Question: Construct a C program to simulate Reader-Writer problem using Semaphores.

Program:

```
1 #include <stdio.h>
2 #include <pthread.h>
3 #include <semaphore.h>
4
5 sem_t mutex, writeBlock;
6 int data = 0, readersCount = 0;
7
8 void *reader(void *arg) {
9     int i = 0;
10    while (i < 10) {
11        sem_wait(&mutex);
12        readersCount++;
13        if (readersCount == 1) {
14            sem_wait(&writeBlock);
15        }
16        sem_post(&mutex);
17
18        printf("Reader reads data: %d\n", data);
19
20        sem_wait(&mutex);
21        readersCount--;
22        if (readersCount == 0) {
23            sem_post(&writeBlock);
24        }
25        sem_post(&mutex);
26        i++;
27    }
28    return NULL;
29 }
30
31 void *writer(void *arg) {
32     int i = 0;
```

Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Writer writes data: 1
Writer writes data: 2
Writer writes data: 3
Writer writes data: 4
Writer writes data: 5
Writer writes data: 6
Writer writes data: 7
Writer writes data: 8
Writer writes data: 9
Writer writes data: 10

...Program finished with exit code 0
Press ENTER to exit console.